ASSIGNMENT 35

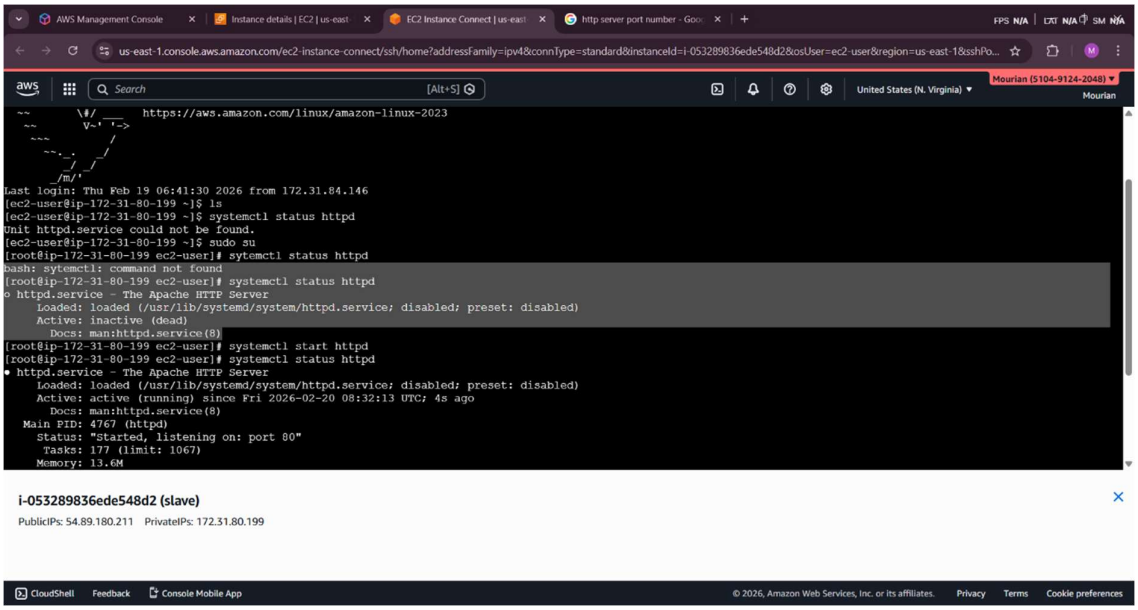
Ansible roles and Vaults

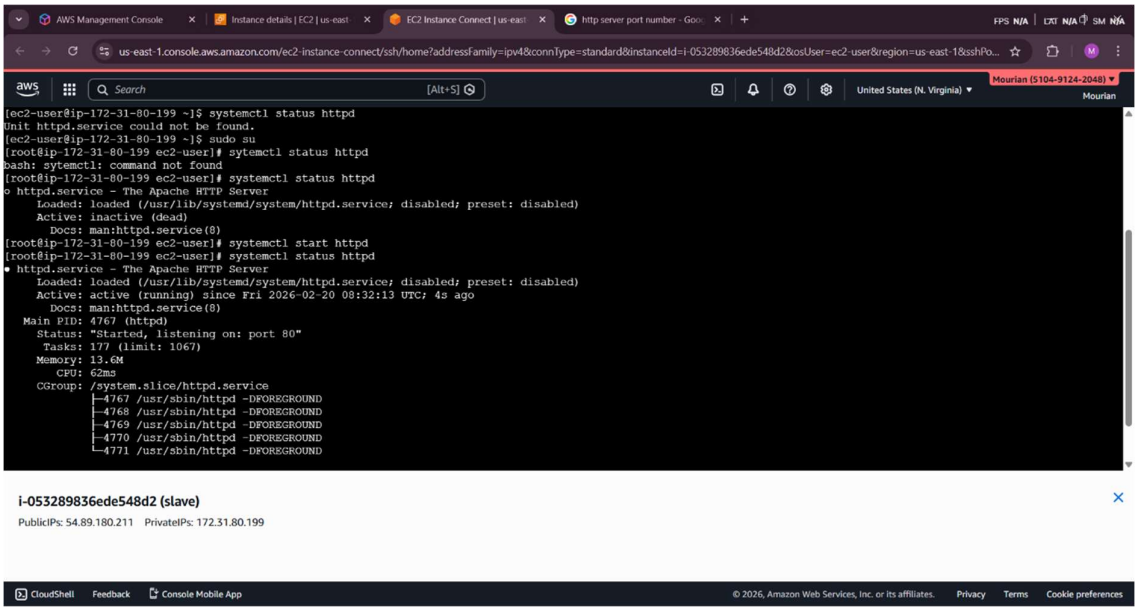
Making a folder of roles as a <myrole>

```
ec2-user@ip-172-31-84-146:~/myrole/tasks

    - name: Verify Node
      command: node -v
      register: node_output
      ignore_errors: yes

    - debug:
        var: node_output.stdout
[ec2-user@ip-172-31-84-146 ~]$ ansible-galaxy init myrole
- Role myrole was created successfully
[ec2-user@ip-172-31-84-146 ~]$ ls
ansible.cfg  inventory.txt  myrole  playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ cd myroles/httpd/tasks/main.yaml
-bash: cd: myroles/httpd/tasks/main.yaml: No such file or directory
[ec2-user@ip-172-31-84-146 ~]$ cd myrole/
[ec2-user@ip-172-31-84-146 myrole]$ ls
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars
[ec2-user@ip-172-31-84-146 myrole]$ cd tasks/
[ec2-user@ip-172-31-84-146 tasks]$ ls
main.yml
[ec2-user@ip-172-31-84-146 tasks]$ nano main.yml
[ec2-user@ip-172-31-84-146 tasks]$ cd ..
[ec2-user@ip-172-31-84-146 myrole]$ cd ..
[ec2-user@ip-172-31-84-146 ~]$ ls
ansible.cfg  inventory.txt  myrole  playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ nano playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ ansible-playbook -i inventory.txt playbook.yaml
ERROR! the role 'httpd' was not found in /home/ec2-user/roles:/home/ec2-user/.ansible/roles:/usr/share/ansible/roles:/etc/ansible/roles:/home/ec2-user

The error appears to be in '/home/ec2-user/playbook.yaml': line 7, column 7, but may
be elsewhere in the file depending on the exact syntax problem.

The offending line appears to be:

  roles:
    - httpd
      ^ here
[ec2-user@ip-172-31-84-146 ~]$ ERROR! the role 'httpd' was not found in /home/ec2-user/roles:/home/ec2-user/.ansible/roles:/usr/share/ansible/roles:/etc/ansible/roles:/home/ec2-user

The error appears to be in '/home/ec2-user/playbook.yaml': line 7, column 7, but may
be elsewhere in the file depending on the exact syntax problem.

The offending line appears to be:

  roles:
    - httpd
      ^ here
^C
[ec2-user@ip-172-31-84-146 ~]$ cd myroles/tasks/main.yaml
```

```
ec2-user@ip-172-31-84-146:~

[ec2-user@ip-172-31-84-146 ~]$ ansible-playbook -i inventory.txt playbook.yaml

PLAY [all] *********************************************************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************************************************
[WARNING]: Platform linux on host 172.31.80.199 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.80.199]
[WARNING]: Platform linux on host 172.31.89.163 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.89.163]

TASK [myrole : Restart service] ************************************************************************************************************************************************************
fatal: [172.31.89.163]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'item' is undefined. 'item' is undefined\n\nThe error appears to be in '/h
ome/ec2-user/myrole/tasks/main.yml': line 3, column 3, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n# tasks file for myro
le\n- name: Restart service\n  ^ here\n"}
fatal: [172.31.80.199]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'item' is undefined. 'item' is undefined\n\nThe error appears to be in '/h
ome/ec2-user/myrole/tasks/main.yml': line 3, column 3, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n# tasks file for myro
le\n- name: Restart service\n  ^ here\n"}

PLAY RECAP *********************************************************************************************************************************************************************************
172.31.80.199              : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
172.31.89.163              : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-84-146 ~]$ nano playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ cd myrole/tasks/
[ec2-user@ip-172-31-84-146 tasks]$ nano main.yml
[ec2-user@ip-172-31-84-146 tasks]$ cd ~
[ec2-user@ip-172-31-84-146 ~]$ ansible-playbook -i inventory.txt playbook.yaml

PLAY [all] *********************************************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************************************************
[WARNING]: Platform linux on host 172.31.80.199 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.80.199]
[WARNING]: Platform linux on host 172.31.89.163 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.89.163]

TASK [myrole : Restart service] ************************************************************************************************************************************************************
changed: [172.31.80.199] => (item=httpd)
changed: [172.31.89.163] => (item=httpd)

PLAY RECAP *********************************************************************************************************************************************************************************
172.31.80.199              : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.89.163              : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-84-146 ~]$
```

Running of httpd in slave machine :





Outputs:

CODE'S Used inside the playbook.yaml & main.yml :

```
ec2-user@ip-172-31-84-146:~/myrole/tasks
le\n- name: Restart service\n  ^ here\n")

PLAY RECAP *********************************************************************************************************************************************************
172.31.80.199              : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
172.31.89.163              : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-84-146 ~]$ nano playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ cd myrole/tasks/
[ec2-user@ip-172-31-84-146 tasks]$ nano main.yml
[ec2-user@ip-172-31-84-146 tasks]$ cd ~
[ec2-user@ip-172-31-84-146 ~]$ ansible-playbook -i inventory.txt playbook.yaml

PLAY [all] *********************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
[WARNING]: Platform linux on host 172.31.80.199 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.80.199]
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meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.89.163]

TASK [myrole : Restart service] ************************************************************************************************************************************
changed: [172.31.80.199] => (item=httpd)
changed: [172.31.89.163] => (item=httpd)

PLAY RECAP *********************************************************************************************************************************************************
172.31.80.199              : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.89.163              : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-84-146 ~]$ ls
ansible.cfg  inventory.txt  myrole  playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ cat playbook.yaml
- hosts: all
  become: yes
  roles:
    - myrole
[ec2-user@ip-172-31-84-146 ~]$ cd myrole/tasks/
[ec2-user@ip-172-31-84-146 tasks]$ cat main.yml
---
# tasks file for myrole
- name: Restart service
  yum:
    name: "{{ item }}"
    state: present
  loop:
    - httpd

[ec2-user@ip-172-31-84-146 tasks]$
```

ec2-user@ip-172-31-84-146:~

```
ok: [172.31.80.199]
[WARNING]: Platform linux on host 172.31.89.163 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.89.163]

TASK [myrole : Restart service] ************************************************************************************************************************************
fatal: [172.31.89.163]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'item' is undefined. 'item' is undefined\n\nThe error appears to be in '/h
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PLAY RECAP *********************************************************************************************************************************************************
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172.31.89.163              : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-84-146 ~]$ nano playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ cd myrole/tasks/
[ec2-user@ip-172-31-84-146 tasks]$ nano main.yml
[ec2-user@ip-172-31-84-146 tasks]$ cd ~
[ec2-user@ip-172-31-84-146 ~]$ ansible-playbook -i inventory.txt playbook.yaml

PLAY [all] *********************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
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172.31.89.163              : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-84-146 ~]$ ls
ansible.cfg  inventory.txt  myrole  playbook.yaml
[ec2-user@ip-172-31-84-146 ~]$ cat playbook.yaml
- hosts: all
  become: yes
  roles:
    - myrole
[ec2-user@ip-172-31-84-146 ~]$
```